

Unit 2

The algebra of vectors and matrices.

Unit Goals

- Be able to explain what is meant by a scalar, a vector, and a matrix.
- Be able to perform fundamental operations (addition and multiplication) with scalars, vectors and matrices.
- Be able to explain the geometric meaning of vectors and their operations.
- Be able to express systems of equations using the simplifying notation of matrices and vectors.
- Be able to perform abstract manipulations of expressions involving vectors and matrices.

Unit 2.1

**The three fundamental objects
of Linear Algebra.**

Three fundamental (mathematical) objects.

Scalar:

- A single number, or a variable representing a single number.
- Typically represented by a lower-case letter (usually italicized if in type): e.g. a, t, x .

Vector:

- An ordered list of scalars.
- Typically represented by a lower-case letter with an arrow over it, or a bold-faced letter if in type: e.g. $\vec{b}, \vec{x}, \vec{u}$ or $\mathbf{b}, \mathbf{x}, \mathbf{u}$.

Three fundamental (mathematical) objects.

Matrix:

- A rectangular array of scalars.
- Typically represented by an upper-case letter; sometimes bold-faced if in type or underlined in hand-written work: e.g. A , U or $\mathbf{A}$, $\mathbf{U}$ or $\underline{A}$, $\underline{U}$.

Scalars

- **Definition:** A scalar is a quantity with only **magnitude** and no **direction**, represented by a single numerical value.
- **Examples:** Temperature, mass, time, speed, and energy are all scalars.
- **Difference from Vectors:** Scalars have only magnitude, while vectors have both **magnitude and direction** (e.g., velocity vs. speed)

Vectors

- **Definition:** A vector is a quantity that has both **magnitude** and **direction**, often represented as an ordered set of numbers.
- Vectors are ordered lists of scalars (e.g. real numbers), typically grouped either by square brackets or parentheses. We can write them horizontally or vertically:

$$\vec{x} = (1, 0, -2), \vec{b} = [2.3, 3.12, 0.1, 8.21], u = \begin{bmatrix} -1 \\ 5 \end{bmatrix}, y = \begin{pmatrix} \sqrt{2} \\ -1 \\ \sqrt{2} \end{pmatrix}$$

Vectors

- **Difference from Scalars:** Unlike scalars, vectors require both magnitude and direction for full definition (e.g., velocity vs. speed).
- **Examples:** Displacement, velocity, force, acceleration, and momentum are all vectors.
- **Mathematical Representation:** Vectors are typically written as $\vec{v}$ or $\mathbf{v}$ and follow vector operations like addition, dot product, and cross product.

Vectors

- We generally use n (and sometimes m) to refer to the number of elements in a vector and refer to an “ n -vector” or an “ n -dimensional vector”:
 - $\vec{u}$ is a 2-vector; $\vec{x}$ and $\vec{y}$ are 3-vectors; $\vec{b}$ is a 4-vector
 - More formally, we write: $\vec{u} \in \mathbb{R}^2$, $\vec{x}, \vec{y} \in \mathbb{R}^3$, $\vec{b} \in \mathbb{R}^4$
- We use v_i to refer to the i^{th} element of a vector $\vec{v}$, with i taking on values between 1 and n (inclusive):

$$x_1 = 1, x_2 = 0, x_3 = -2, b_4 = 8.21, u_1 = -1, u_2 = 5, y_1 = y_3 = \sqrt{2}$$

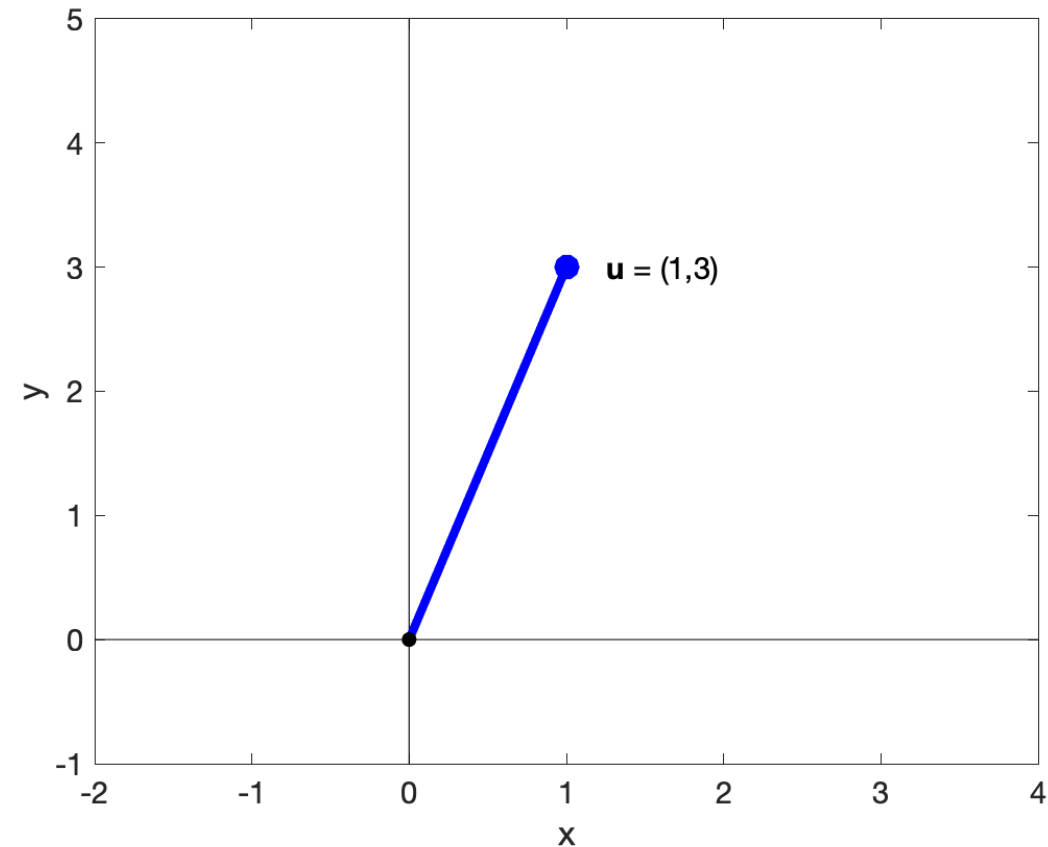
Spatial vectors and points in space

- Points in space can be represented by ordered pairs or triplets of real numbers, making vectors a natural way to represent these points.
- A point in space can also be described as a directed line segment from the origin to that point, which can be represented as a vector.
- These geometric interpretations help in understanding vector operations and the fundamental rules governing them.
- However, vectors are not limited to spatial locations: they can represent many other concepts, such as forces, velocities, and abstract mathematical quantities.

Spatial vectors and points in space

Example:

- The vector $(1,3)$ may either be thought of as the **point** located at $x=1$, $y=3$ in the x - y plane, or
- as the directed **line segment** that starts at the origin $(x=0, y=0)$ and ends at $(x=1, y=3)$.



Matrices

- Matrices are ordered rectangular **arrays of scalars** (e.g. real numbers), typically grouped by square brackets or by parentheses:

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, V = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 3 \\ 1 & 3 & 0 \end{pmatrix}, U = \begin{bmatrix} 1.1 & -0.2 & 2.7 & -0.2 \\ 3.9 & 0.7 & 2.1 & 1.0 \end{bmatrix}, \underline{M} = \begin{bmatrix} \sqrt{3} & 1 \\ \sqrt{3} & 0 \\ \sqrt{3} & -1 \end{bmatrix}$$

- We often use m to refer to the number of rows in a matrix and n to refer to the number of columns; we then refer to an “ m -by- n matrix” or an “ $m \times n$ matrix”. **Note, the number of rows always comes first:**
 - A is a 2-by-2 matrix; V is a 3x3 matrix; U is a 2x4 matrix; M is a 3x2 matrix
 - More formally, we write: $A \in \mathbb{R}^{2 \times 2}$, $V \in \mathbb{R}^{3 \times 3}$, $U \in \mathbb{R}^{2 \times 3}$, $M \in \mathbb{R}^{3 \times 2}$

Matrices

- We use a_{ij} to refer to the element in the i^{th} row and j^{th} column of a matrix A , with i taking on values between 1 and m , and j taking on values between 1 and n :

$$a_{11} = 1, a_{12} = 2, a_{21} = 3, a_{22} = 4,$$

$$u_{11} = 0.1, u_{12} = 0.2, u_{21} = -0.1, u_{22} = -0.2,$$

$$m_{11} = m_{21} = m_{31} = \sqrt{3}$$

Matrices

- Matrices are fundamental tools used to represent various real-world concepts.
- Throughout this class, we will explore their applications in:
 - **Coefficients** of a linear system;
 - Physical or optical transformation;
 - **Networks** and **graph**;
 - Partial **derivatives** functions;
 - **Covariances** structures in dataset.

Unit 2.2

Addition and Scalar Multiplication of Vectors and Matrices

Three fundamental operations.

Our exploration with the three fundamental objects (**scalars, vectors, and matrices**) would lack practical utility without using the objects for three core operations:

1. **Addition**: We can add any two objects of the same type and size by adding the corresponding elements.
2. **Scalar multiplication**: We multiply every element in the object by the same scalar.
3. **Matrix-matrix, matrix-vector and vector-vector multiplication**: We can multiply certain vectors and matrices according to specific rules.

Addition

The addition involving vectors and matrices is a very straight forward process:

Vectors: $\vec{x} + \vec{y} = \vec{z} \xrightarrow{\text{implies}} z_i = x_i + y_i, \forall i \text{ (for all } i)$
 $(1,0,2) + (-1,5,2) = (1 - 1, 0 + 5, 2 + 2) = (0,5,4)$

Matrices: $A + B = C \xrightarrow{\text{implies}} c_{ij} = a_{ij} + b_{ij}, \forall i, j \text{ (for all } i, j \text{ pairs)}$
$$\begin{bmatrix} 3 & -2 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} -2 & -2 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 3 - 2 & -2 - 2 \\ 2 + 4 & 1 + 7 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 6 & 8 \end{bmatrix}$$

Addition

Note that:

- **Objects of different types can not be added.**
- **Objects of the same type, but of different sizes, can not be added.**

Scalar multiplication

Scalar multiplication is the familiar process of **repeated addition** (e.g., $3 \times 5 = 5+5+5$)

It applies equally to vectors and matrices: every element is multiplied by the scalar value.

Vectors:
$$k\vec{u} = \vec{v} \xrightarrow{\text{implies}} v_i = ku_i, \forall i \text{ (for all } i)$$
$$7(1, -2, 4) = (7(1), 7(-2), 7(4)) = (7, -14, 28)$$

Matrices:
$$kA = C \xrightarrow{\text{implies}} c_{ij} = ka_{ij}, \forall i, j \text{ (for all } i, j \text{ pairs)}$$
$$(0.5) \begin{bmatrix} 3 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} (0.5)(3) & (0.5)(-2) \\ (0.5)(2) & (0.5)(1) \end{bmatrix} = \begin{bmatrix} 1.5 & -1 \\ 1 & 0.5 \end{bmatrix}$$

Linear combinations

- The concept of a linear combination is the core ideas in linear algebra. We now have the tools to define it!
- **A linear combination of a set of vectors or matrices is simply a sum of scalar multiples of the members of the set.**

$$a\vec{x} + b\vec{y} = \vec{z} \quad \rightarrow \vec{z} \text{ is a linear combination of } \vec{x} \text{ and } \vec{y}.$$

$$a\vec{x} + b\vec{y} + c\vec{z} = \vec{w} \quad \rightarrow \vec{w} \text{ is a linear combination of } \vec{x}, \vec{y} \text{ and } \vec{z}.$$

$$aA + bB + cC + dD = M \quad \rightarrow M \text{ is a linear combination of } A, B, C \text{ and } D.$$

$$\sum_{i=1}^m a_i \vec{x}_i = \vec{y} \quad \rightarrow \vec{y} \text{ is a linear combination of the set of vectors } \{\vec{x}_i\}.$$

$$\sum_{i=1}^m a_i X_i = Y \quad \rightarrow Y \text{ is a linear combination of the set of matrices } \{X_i\}.$$

The geometry of addition and scalar multiplication of spatial vectors.

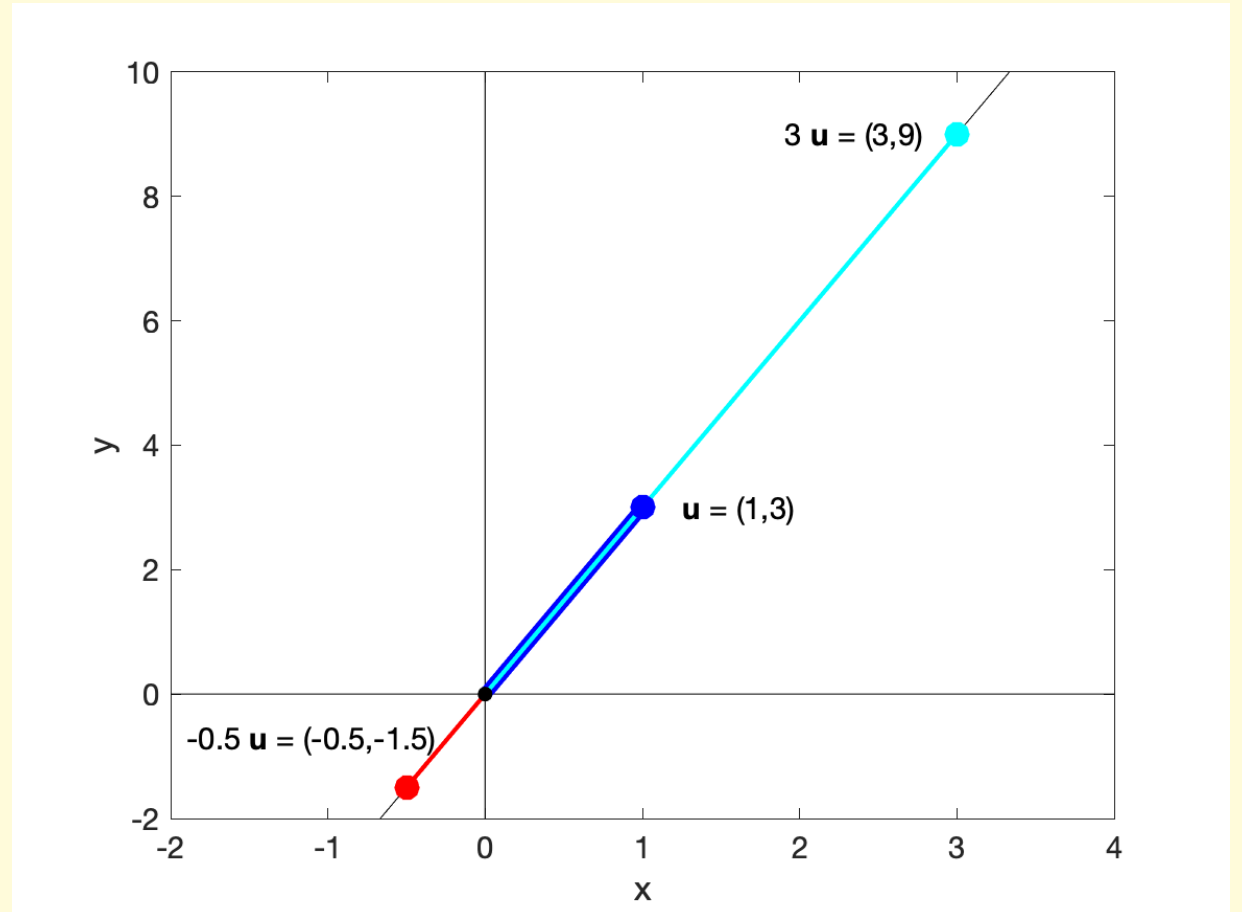
- If we think of vectors (especially 2D and 3D vectors) as points in space, we can explain addition and scalar multiplication geometrically:
 - Scalar multiplication by a **positive** value changes the length of the vector.
 - Scalar multiplication by a **negative** value reverses the direction (and changes the length) of the vector.
 - Vector addition is done “head to tail” and follows the “**parallelogram**” rule.
 - Vector subtraction is the same as adding a scalar multiple of -1. It can be represented by directly connecting the points (vector tips).

The geometry of addition and scalar multiplication of spatial vectors.

- The **parallelogram rule** is a method used in vector addition to determine the resultant of two vectors.
- It states that if two vectors are represented as adjacent sides of a parallelogram, then their sum is given by the diagonal of the parallelogram starting from the same point.

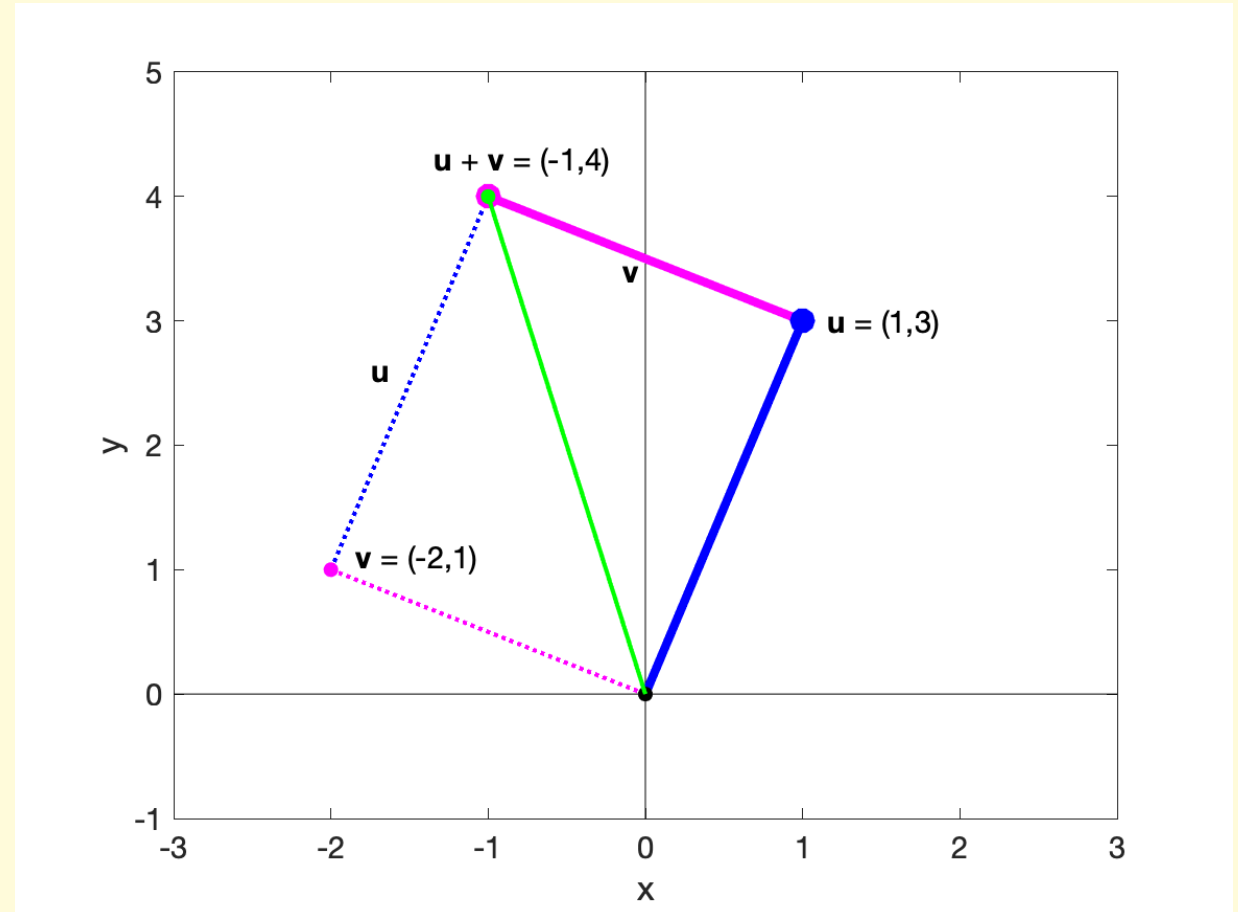
The geometry of addition and scalar multiplication of spatial vectors.

- Scalar multiplication by a positive value changes the length of the vector.
- Scalar multiplication by a negative value reverses the direction (and changes the length) of the vector.



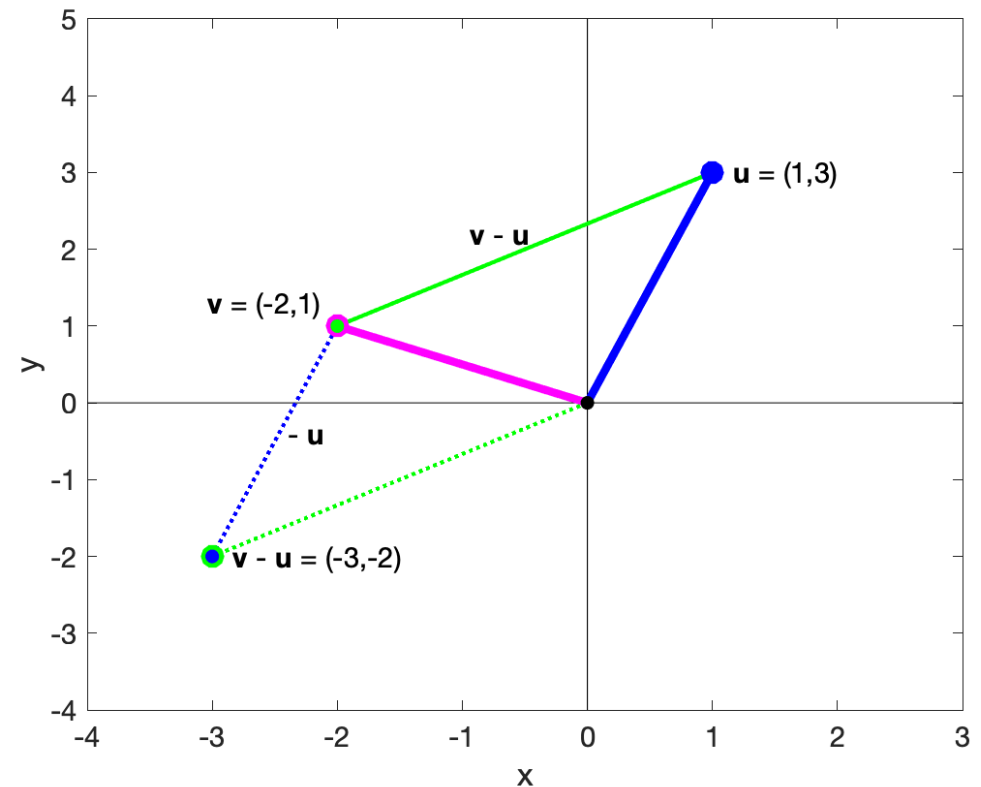
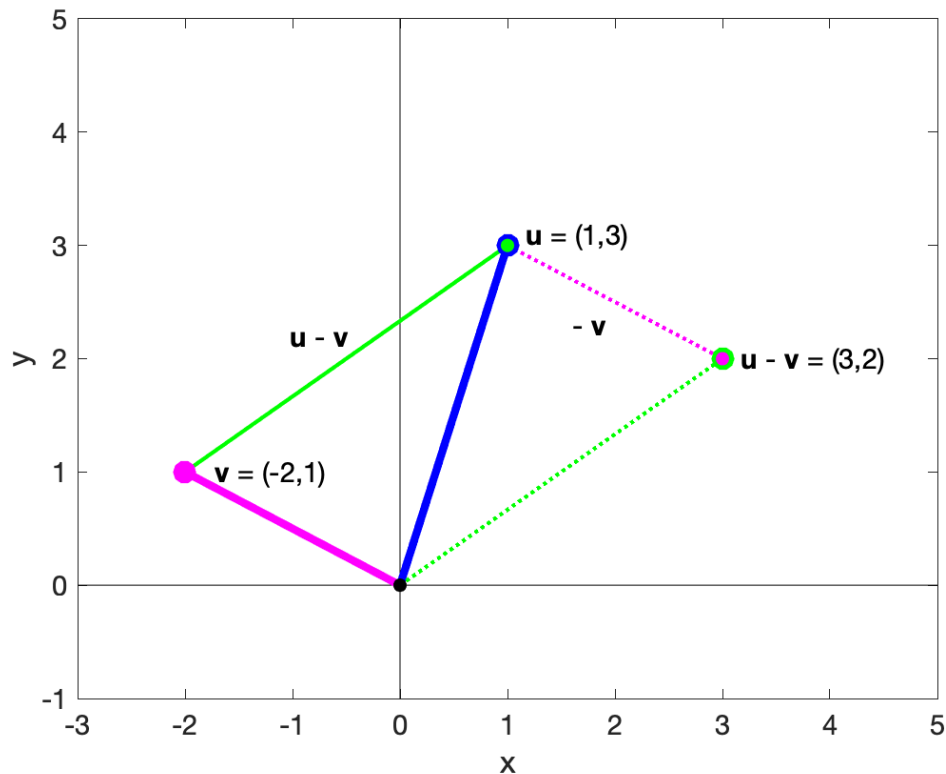
The geometry of addition and scalar multiplication of spatial vectors.

- Vector addition is done “head to tail” and follows the “**parallelogram**” rule.



The geometry of addition and scalar multiplication of spatial vectors.

- Vector subtraction is the same as adding a scalar multiple of -1. It can be represented by directly connecting the points (vector tips).



Unit 2.2.1

Worked problems in addition and scalar multiplication of vectors and matrices.

Operations with vectors.

- Compute the following operations involving vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = (1, 0, 2) \qquad \vec{v} = (1, -1, -1)$$

(a) $2\vec{u} + \vec{v}$

(b) $\vec{u} - 2\vec{v}$

Operations with vectors.

- Compute the following operations involving vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = (1, 0, 2) \qquad \vec{v} = (1, -1, -1)$$

$$\begin{aligned} \text{(a)} \quad 2\vec{u} + \vec{v} &= 2(1, 0, 2) + (1, -1, -1) \\ &= (2, 0, 4) + (1, -1, -1) \\ &= (3, -1, 3) \end{aligned}$$

Operations with vectors.

- Compute the following operations involving vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = (1, 0, 2) \qquad \vec{v} = (1, -1, -1)$$

$$\begin{aligned} \text{(b)} \quad \vec{u} - 2\vec{v} &= (1, 0, 2) - 2(1, -1, -1) \\ &= (1, 0, 2) - (2, -2, -2) \\ &= (-1, 2, 4) \end{aligned}$$

Operations with matrices.

- Compute the following operations involving matrices A , B and C :

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

(a) $2A + 3B$

(b) $2(A - B) + C$

Operations with matrices.

- Compute the following operations involving matrices A , B and C :

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

$$\begin{aligned} \text{(a)} \quad 2A + 3B &= 2 \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} + 3 \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & -6 \\ 3 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -6 \\ 5 & 7 \end{bmatrix} \end{aligned}$$

Operations with matrices.

- Compute the following operations involving matrices A , B and C :

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, B = \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix}, C = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

$$\begin{aligned} \text{(b)} \quad 2(A - B) + C &= 2 \left(\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix} \right) + \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} \\ &= 2 \left(\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \right) + \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 7 \\ -1 & 0 \end{bmatrix} \end{aligned}$$

Linear combinations.

- Find linear combinations of $\vec{u}$ and $\vec{v}$ that equal the given $\vec{w}$:

$$\vec{u} = (1, 0, 2) \qquad \vec{v} = (1, -1, -1)$$

(a) $\vec{w} = (2, -1, 1)$

(b) $\vec{w} = (2, -3, -5)$

(c) $\vec{w} = (4, -2, 0)$

Linear combinations.

- Find linear combinations of $\vec{u}$ and $\vec{v}$ that equal the given $\vec{w}$:

$$\vec{u} = (1,0,2) \quad \vec{v} = (1,-1,-1)$$

- We want to find a and b such that $\vec{w} = a\vec{u} + b\vec{v}$.
- We note that the middle term of $a\vec{u}$ will always be 0. Therefore, the middle term in $\vec{w}$ must come from $b\vec{v}$:

$$\begin{aligned} a\vec{u} + b\vec{v} &= a(1,0,2) + b(1,-1,-1) \\ &= (a, 0, 2a) + (b, -b, -b) \\ &= (a + b, -b, 2a - b) \end{aligned}$$

Linear combinations.

- Find linear combinations of $\vec{u}$ and $\vec{v}$ that equal the given $\vec{w}$:

$$\vec{u} = (1, 0, 2) \quad \vec{v} = (1, -1, -1)$$

$$(a) \quad \vec{w} = (2, -1, 1) = (a + b, -b, 2a - b)$$

$$\text{Second term:} \quad b = 1$$

$$\text{First term:} \quad 2 = a + 1 \rightarrow a = 1$$

$$\text{Third term:} \quad 1 = 2a - 1 \rightarrow a = 1$$

$$(2, -1, 1) = (1, 0, 2) + (1, -1, -1)$$

Linear combinations.

- Find linear combinations of $\vec{u}$ and $\vec{v}$ that equal the given $\vec{w}$:

$$\vec{u} = (1, 0, 2) \quad \vec{v} = (1, -1, -1)$$

$$(b) \quad \vec{w} = (2, -3, -5) = (a + b, -b, 2a - b)$$

$$\text{Second term:} \quad b = 3$$

$$\text{First term:} \quad 2 = a + 3 \rightarrow a = -1$$

$$\text{Third term:} \quad -5 = 2a - 3 \rightarrow a = -1$$

$$(2, -3, -5) = -1(1, 0, 2) + 3(1, -1, -1)$$

Linear combinations.

- Find linear combinations of $\vec{u}$ and $\vec{v}$ that equal the given $\vec{w}$:

$$\vec{u} = (1, 0, 2) \quad \vec{v} = (1, -1, -1)$$

(c) $\vec{w} = (4, -2, 0) = (a + b, -b, 2a - b)$

Second term: $b = 2$

First term: $4 = a + 2 \rightarrow a = 2$

Third term: $0 = 2a - 2 \rightarrow a = 1$

There is no way to write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$.

Unit 2.3

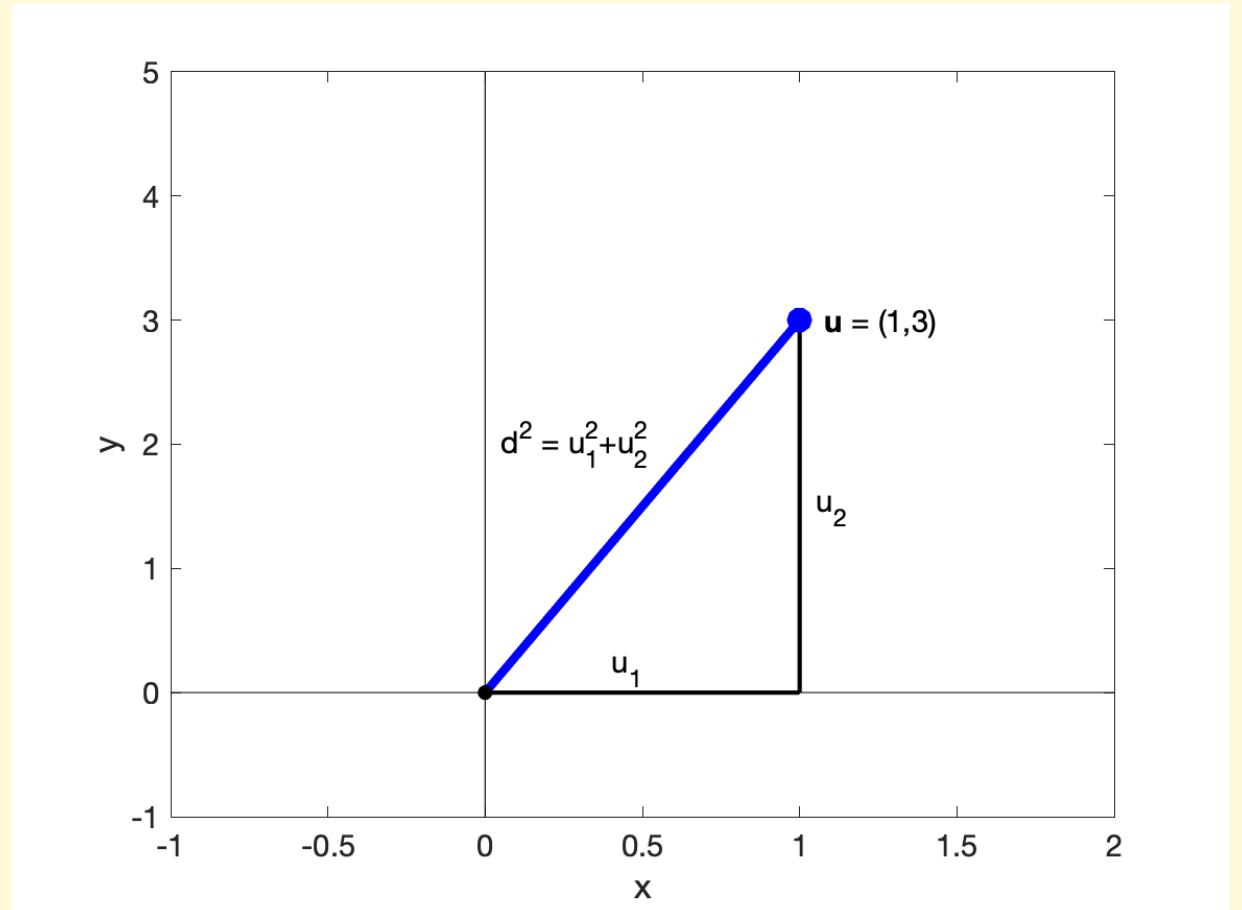
Matrix multiplication

More geometry of spatial vectors

- The length of vector is given by the **square root** of the sum of the individual terms.
- We often refer to this length as the “**norm**” of the vector, and use the notation $d = \|\vec{u}\|$:

$$\|\vec{u}\|^2 = \sum_{i=1}^n (u_i)^2$$

- **2-D:** $\|\vec{u}\| = \sqrt{(u_1)^2 + (u_2)^2}$
- **3-D:** $\|\vec{u}\| = \sqrt{(u_1)^2 + (u_2)^2 + (u_3)^2}$



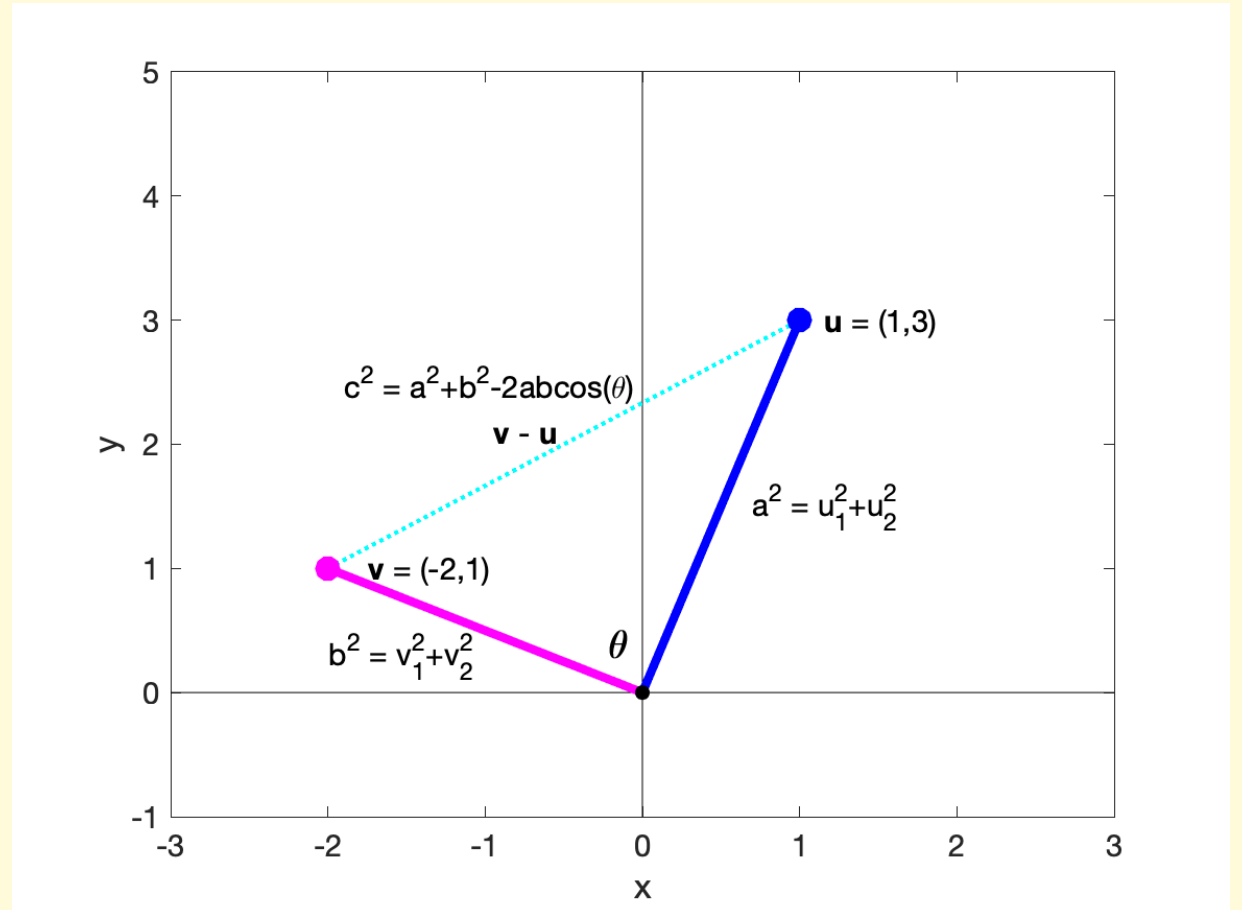
More geometry of spatial vectors

- The angle between two vectors can be given by the Cosine Law.

$$\|\vec{v} - \vec{u}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\|\vec{u}\|\|\vec{v}\|\cos\theta$$

$$2\|\vec{u}\|\|\vec{v}\|\cos\theta = \|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{v} - \vec{u}\|^2$$

$$\cos\theta = \frac{\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{v} - \vec{u}\|^2}{2\|\vec{u}\|\|\vec{v}\|}$$



An interesting result.

$$\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{v} - \vec{u}\|^2$$

$$= \sum_{i=1}^n (u_i)^2 + \sum_{i=1}^n (v_i)^2 - \sum_{i=1}^n (v_i - u_i)^2$$

$$= \sum_{i=1}^n (u_i)^2 + \sum_{i=1}^n (v_i)^2 - \left(\sum_{i=1}^n (v_i)^2 + \sum_{i=1}^n (-2v_i u_i) + \sum_{i=1}^n (-u_i)^2 \right)$$

$$= \sum_{i=1}^n (u_i)^2 + \sum_{i=1}^n (v_i)^2 - \sum_{i=1}^n (v_i)^2 + \sum_{i=1}^n (2v_i u_i) - \sum_{i=1}^n (u_i)^2$$

$$= 2 \sum_{i=1}^n (v_i u_i)$$

$$\cos \theta = \frac{\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{v} - \vec{u}\|^2}{2\|\vec{u}\|\|\vec{v}\|}$$

$$\cos \theta = \frac{\sum_{i=1}^n u_i v_i}{\|\vec{u}\|\|\vec{v}\|}$$

$$\cos \theta = \frac{\sum_{i=1}^n u_i v_i}{\sqrt{\sum_{i=1}^n (u_i)^2} \sqrt{\sum_{i=1}^n (v_i)^2}}$$

- We have three sums of the same form!

The dot product.

- We define the dot product, or scalar product, of two vectors to be the **sum of the products of corresponding entries**:

$$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n$$
$$\vec{u} \cdot \vec{v} = \sum_{i=1}^n u_i v_i$$

Notes:

- The dot product **gives a scalar** as the results.
- The two vectors must be of the same size (dimension).

The geometry of the dot product.

- The length (norm) of a vector is the square root of the dot product of the vector with itself:

$$\|\vec{u}\|^2 = \sum_{i=1}^n (u_i)^2 = \vec{u} \cdot \vec{u}$$
$$\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}}$$

- The cosine of the angle between two vectors is the dot product of the two vectors divided by their lengths:

$$\cos \theta = \frac{\sum_{i=1}^n u_i v_i}{\|\vec{u}\| \|\vec{v}\|} = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}$$

- This property make the dot product very useful in many applications.

Matrix multiplication

We have seen that we can define a product between two vectors, the dot product. We will now generalize this idea to ~~products~~ **perform multiplication** between matrices.

- The i, j^{th} element of the product, AB , is obtained by walking along row i of A and column j of B , multiplying corresponding elements, and adding them all up:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 6 & 0 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} (1)(4) + (0)(6) + (-1)(5) & (1)(2) + (0)(0) + (-1)(-2) \\ (2)(4) + (1)(6) + (3)(5) & (2)(2) + (1)(0) + (3)(-2) \end{bmatrix}$$
$$= \begin{bmatrix} 4 + 0 - 5 & 2 + 0 + 2 \\ 8 + 6 + 15 & 4 + 0 - 6 \end{bmatrix} = \begin{bmatrix} -1 & 4 \\ 29 & -2 \end{bmatrix}$$

Matrix multiplication

- The rows of A must be the same length as the columns of B
- Let A be $(m \times r)$ and B be $(r \times n)$; then $C = AB$ must be $(m \times n)$.
- The inner dimensions of A and B must match, and this dimension “disappears” in the product

Matrix multiplication

- We are taking the dot product of row i of A and column j of B , for each entry c_{ij} in the product:

$$C = AB \rightarrow c_{ij} = \vec{a}_i^R \cdot \vec{b}_j^C$$

$$A = \begin{bmatrix} \vec{a}_1^R \\ \vec{a}_2^R \\ \vdots \\ \vec{a}_m^R \end{bmatrix}, \quad B = [\vec{b}_1^C \quad \vec{b}_2^C \quad \cdots \quad \vec{b}_n^C]$$

Vectors as matrices.

- Matrices that have either a single row or a single column look very much like vectors, and we will see they behave nearly identically.
- This leads to the distinction of a row vector (a $1 \times n$ matrix) and a column vector (an n -by-1 matrix):

$$\vec{u}_R = [-1 \quad 4 \quad 2], \quad \vec{u}_C = \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}$$

- The transpose is an operation that converts the rows of a matrix into into columns, and vice versa:

$$\vec{u}_C = (\vec{u}_R)^T, \quad \vec{u}_R = (\vec{u}_C)^T$$

The dot product as matrix multiplication.

- We can write the **dot product** as a matrix product.
- We will choose a standard of column vectors; all vectors will be assumed to be columns, and we use the transpose if we want a row vector:

$$\vec{u} \in \mathbb{R}^n \leftrightarrow \vec{u} \in \mathbb{R}^{n \times 1},$$

$$\vec{u}^T \in \mathbb{R}^{1 \times n} \quad \text{or}$$

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \vec{u}^T = [u_1 \quad u_2 \quad \cdots \quad u_n]$$

The dot product as matrix multiplication.

- Subsequently:

$$\begin{aligned}\vec{v}^T \vec{u} &= [v_1 \quad v_2 \quad \cdots \quad v_n] \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \\ &= u_1 v_1 + u_2 v_2 + \cdots + u_n v_n \\ &= \vec{u} \cdot \vec{v}\end{aligned}$$

The matrix transpose

- The transpose is an operation that converts the rows of a matrix into columns, and vice versa.
- For row and column vectors, this is trivial, but we can generalize the operation to any matrix.

$$A = \begin{bmatrix} \vec{a}_1^R \\ \vec{a}_2^R \\ \vdots \\ \vec{a}_m^R \end{bmatrix} \rightarrow A^T = [\vec{a}_1^C \quad \vec{a}_2^C \quad \cdots \quad \vec{a}_m^C], \quad B = [\vec{b}_1^C \quad \vec{b}_2^C \quad \cdots \quad \vec{b}_n^C] \rightarrow B^T = \begin{bmatrix} \vec{b}_1^R \\ \vec{b}_2^R \\ \vdots \\ \vec{b}_n^R \end{bmatrix}$$

where: $\vec{a}_i^C = (\vec{a}_i^R)^T$ and $\vec{b}_i^R = (\vec{b}_i^C)^T$

- The transpose of an m-by-n matrix is an n-by-m matrix.

Special case – Identity matrices.

- A special **square matrix** is one that has 1's for every entry on the diagonal, and 0's everywhere else
- We can define this for any size square matrix, and refer to these as **identity matrix, I** :

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I_n = \begin{bmatrix} 1 & 0 & 0 & & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & & 0 & 0 \\ & \vdots & & \ddots & & \vdots \\ 0 & 0 & 0 & & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{bmatrix}$$

Special case – Identity matrices.

- Multiplication of any matrix by the identity matrix (as long as the product is well-defined) does not change the matrix!

$$IA = A \qquad BI = B$$

Special case – Matrix inverses.

- The identity matrix behaves much like the **scalar 1**.
- It is the **multiplicative** identity.
- For all scalars other than 0, we can define a multiplicative inverse: a unique number that we multiply by to get 1:

$$xy = 1 \rightarrow y = \frac{1}{x} = x^{-1}, x = \frac{1}{y} = y^{-1}$$

Special case – Zero matrices.

- Another special matrix is one that has 0's everywhere; we can define this for any size matrix (square or non-square), and refer to these as ~~identity~~ **zero** matrices, **0**:

$$0_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad 0_{3,3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad 0_{3,2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad 0_{2,4} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- Note that we very often just use 0, without a subscript; it is then assumed that we are using the “correct” size zero matrix for the problem.

Special case – Zero matrices.

- Addition of the zero matrix of appropriate size to any other matrix gives the same matrix:

$$\mathbf{0} + A = A + \mathbf{0} = A$$

- Multiplication of any matrix by a zero matrix gives another zero matrix. However, the **size may change!**

$$A\mathbf{0} = \mathbf{0}, \mathbf{0}A = \mathbf{0}$$

Matrix-vector products.

- When we think of a vector as an n -by-1 *column* matrix, the rules of matrix multiplication means that the product $A\vec{x} = \vec{b}$ will be well-defined for any $(m \times n)$ matrix A and any $(n \times 1)$ vector $\vec{x}$, and the result will be an $(m \times 1)$ vector $\vec{b}$.

$$(m \times n)(n \times 1) \rightarrow (m \times 1)$$

Matrix-vector products.

- Similarly, for the transpose of such a vector (a 1-by- n *row* matrix), the rules of matrix multiplication means that the product $\vec{x}^T A = \vec{b}^T$ will be well-defined for any $(m \times n)$ matrix A and any $(1 \times m)$ vector $\vec{x}^T$, and the result will be a $(1 \times n)$ vector $\vec{b}^T$.

$$(1 \times m)(m \times n) \rightarrow (1 \times n)$$

Matrix-vector products.

- A very special thing happens when A is a matrix of constants and $\vec{x}$ is a vector of variables.

$$\begin{bmatrix} 2 & 3 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + 3y \\ x - 2y \end{bmatrix}$$

- The result looks just like the left-hand side of a system of equations:

$$\begin{aligned} 2x + 3y &= 7 \\ x - 2y &= -5 \end{aligned}$$

- The right-hand side is just a pair of numbers that we can represent by a vector:

$$\begin{bmatrix} 2x + 3y \\ x - 2y \end{bmatrix} = \begin{bmatrix} 7 \\ -5 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ -5 \end{bmatrix} \rightarrow A\vec{x} = \vec{b}$$

Matrix-vector products.

- A very special thing happens when A is a matrix of constants and $\vec{x}$ is a vector of variables:

$$A\vec{x} = \vec{b} \quad \text{represents a system of linear equations.}$$

- The number of rows in A corresponds to the number of equations in the system.
- The number of columns in A corresponds to the number of variables.
- The product $A\vec{x}$ corresponds to the left-hand side (the side with variables).
- $\vec{b}$ corresponds to the constants on the right-hand side.

Unit 2.3.1

**Worked problems in vector and
matrix multiplication.**

Scalar products of vectors.

- Compute the following operations involving vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = (1, 0, 2) \quad \vec{v} = (1, -1, -1)$$

- (a) $\vec{u} \cdot \vec{v}$
- (b) $\|\vec{v}\|$
- (c) $(2\vec{u} + \vec{v}) \cdot (\vec{u} - 2\vec{v})$
- (d) $\|\vec{v} - \vec{u}\|$

Scalar products of vectors.

- Compute the following operations involving vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = (1, 0, 2) \quad \vec{v} = (1, -1, -1)$$

(a) $\vec{u} \cdot \vec{v}$

$$\begin{aligned}\vec{u} \cdot \vec{v} &= (1, 0, 2) \cdot (1, -1, -1) \\ &= (1)(1) + (0)(-1) + (2)(-1) \\ &= 1 + 0 - 2 \\ &= -1\end{aligned}$$

Scalar products of vectors.

- Compute the following operations involving vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = (1, 0, 2) \quad \vec{v} = (1, -1, -1)$$

(b) $\|\vec{v}\|$

$$\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}}$$

$$\vec{v} \cdot \vec{v} = (1)^2 + (-1)^2 + (-1)^2$$

$$= 1 + 1 + 1$$

$$= 3$$

$$\|\vec{v}\| = \sqrt{3}$$

Scalar products of vectors.

- Compute the following operations involving vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = (1,0,2) \quad \vec{v} = (1,-1,-1)$$

(c) $(2\vec{u} + \vec{v}) \cdot (\vec{u} - 2\vec{v})$

$$(2\vec{u} + \vec{v}) = 2(1,0,2) + (1,-1,-1) = (2+1, 0-1, 4-1) = (3,-1,3)$$

$$(\vec{u} - 2\vec{v}) = (1,0,2) - 2(1,-1,-1) = (1-2, 0+2, 2+2) = (-1,2,4)$$

$$\begin{aligned}(2\vec{u} + \vec{v}) \cdot (\vec{u} - 2\vec{v}) &= (3,-1,3) \cdot (-1,2,4) \\ &= (3)(-1) + (-1)(2) + (3)(4) \\ &= -3 - 2 + 12 \\ &= 7\end{aligned}$$

Scalar products of vectors.

- Compute the following operations involving vectors $\vec{u}$ and $\vec{v}$:

$$\vec{u} = (1, 0, 2) \quad \vec{v} = (1, -1, -1)$$

(d) $\|\vec{v} - \vec{u}\|$

$$(\vec{v} - \vec{u}) = (1, -1, -1) - (1, 0, 2) = (1 - 1, -1 - 0, -1 - 2) = (0, -1, -3)$$

$$(\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}) = (0)^2 + (-1)^2 + (-3)^2$$

$$= 0 + 1 + 9$$

$$= 10$$

$$\|\vec{v} - \vec{u}\| = \sqrt{(\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u})} = \sqrt{10}$$

Matrix products.

- Compute the following matrix products, or explain why it is not possible to do so:

(a) AB

(b) CB

(c) ABC

(d) ACB

where:

$$A = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & 1 \end{bmatrix} \quad C = \begin{bmatrix} -1 & 3 \\ 1 & 0 \\ -1 & 3 \end{bmatrix}$$

Matrix products.

- Compute the following matrix products, or explain why it is not possible to do so:

(a) AB where: $A = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & 1 \end{bmatrix}$ $C = \begin{bmatrix} -1 & 3 \\ 1 & 0 \\ -1 & 3 \end{bmatrix}$

$$\begin{aligned} AB &= \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2(1) - 1(2) & 2(0) - 1(2) & 2(-1) - 1(1) \\ 0(1) + 1(2) & 0(0) + 1(2) & 0(-1) + 1(1) \end{bmatrix} \\ &= \begin{bmatrix} 2 - 2 & 0 - 2 & -2 - 1 \\ 0 + 2 & 0 + 2 & 0 + 1 \end{bmatrix} = \begin{bmatrix} 0 & -2 & -3 \\ 2 & 2 & 1 \end{bmatrix} \end{aligned}$$

Matrix products.

- Compute the following matrix products, or explain why it is not possible to do so:

(b) CB where: $A = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & 1 \end{bmatrix}$ $C = \begin{bmatrix} -1 & 3 \\ 1 & 0 \\ -1 & 3 \end{bmatrix}$

$$CB = \begin{bmatrix} -1 & 3 \\ 1 & 0 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1(1) + 3(2) & -1(0) + 3(2) & -1(-1) + 3(1) \\ 1(1) + 0(2) & 1(0) + 0(2) & 1(-1) + 0(1) \\ -1(1) + 3(2) & -1(0) + 3(2) & -1(-1) + 3(1) \end{bmatrix}$$

$$= \begin{bmatrix} -1 + 6 & 0 + 6 & 1 + 3 \\ 1 + 0 & 0 + 0 & -1 + 0 \\ -1 + 6 & 0 + 6 & 1 + 3 \end{bmatrix} = \begin{bmatrix} 5 & 6 & 4 \\ 1 & 0 & -1 \\ 5 & 6 & 4 \end{bmatrix}$$

Matrix products.

- Compute the following matrix products, or explain why it is not possible to do so:

(c) ABC where: $A = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & 1 \end{bmatrix}$ $C = \begin{bmatrix} -1 & 3 \\ 1 & 0 \\ -1 & 3 \end{bmatrix}$

Recall: $AB = \begin{bmatrix} 0 & -2 & -3 \\ 2 & 2 & 0 \end{bmatrix}$

$$\begin{aligned} ABC &= A(BC) = \begin{bmatrix} 0 & -2 & -3 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 1 & 0 \\ -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 0(-1) - 2(1) - 3(-1) & 0(3) - 2(0) - 3(3) \\ 2(-1) + 2(1) + 0(-1) & 2(3) + 2(0) + 0(3) \end{bmatrix} \\ &= \begin{bmatrix} 0 - 2 + 3 & 0 - 0 - 9 \\ -2 + 2 + 0 & 6 + 0 + 0 \end{bmatrix} = \begin{bmatrix} 1 & -9 \\ 0 & 6 \end{bmatrix} \end{aligned}$$

Matrix products.

- Compute the following matrix products, or explain why it is not possible to do so:

(d) ACB where: $A = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & 1 \end{bmatrix}$ $C = \begin{bmatrix} -1 & 3 \\ 1 & 0 \\ -1 & 3 \end{bmatrix}$

Recall: $CB = \begin{bmatrix} 5 & 6 & 4 \\ 1 & 0 & -1 \\ 5 & 6 & 4 \end{bmatrix}$

$$ACB = A(CB) = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 6 & 4 \\ 1 & 0 & -1 \\ 5 & 6 & 4 \end{bmatrix}$$

- This is not a well-defined expression, as the inner dimensions do not match.

$$A(CB) \rightarrow (2 \times 2)(3 \times 3)$$

$$ACB \rightarrow (2 \times 2)(3 \times 2)(2 \times 3)$$

Unit 2.3.2

**Worked problems expressing
linear systems as matrix-vector
products.**

An industrial production problem.

- An oil company has a number classes of refinery with different production outputs per barrel of crude oil:
 - Type 1 produces 15 gallons of gasoline, 5 gallons of kerosene (jet fuel), 10 gallons of diesel, 5 gallons of (heavy) fuel oil and 5 gallons of asphalt.
 - Type 2 produces 20 gallons of gasoline, 10 gallons of kerosene, and 10 gallons of diesel, with no other products.
 - Type 3 produces 10 gallons of gasoline, 5 gallons of kerosene, 5 gallons of diesel, 10 gallons of heavy fuel oil and 5 gallons of asphalt.
- How many barrels should be processed by each class in order to produce 80 M gallons of gasoline, 30 M gallons of kerosene, 40 M gallons of diesel, 45 M gallons of fuel oil and 30 M gallons of asphalt?
- Let x_i be the number of barrels (in millions) processed by Type “i” refineries, $i=1,2,3$.
- Gasoline: $15x_1 + 20x_2 + 10x_3 = 80$
- Kerosene: $5x_1 + 10x_2 + 5x_3 = 30$
- Diesel: $10x_1 + 10x_2 + 5x_3 = 40$
- Fuel Oil: $5x_1 + 0x_2 + 10x_3 = 45$
- Asphalt: $5x_1 + 0x_2 + 5x_3 = 30$
- We would like to find the values of x_1 , x_2 and x_3 that simultaneously make all equations true.
- Translate this problem into the language of matrices and vectors.

An industrial production problem.

Variables are the same in every equation → Vector.

Coefficients vary in both row and column → Matrix.

$$\begin{bmatrix} 15x_1 & +20x_2 & +10x_3 \\ 5x_1 & +10x_2 & +5x_3 \\ 10x_1 & +10x_2 & +5x_3 \\ 5x_1 & +0x_2 & +10x_3 \\ 5x_1 & +0x_2 & +5x_3 \end{bmatrix} = \begin{bmatrix} 15 & 20 & 10 \\ 5 & 10 & 5 \\ 10 & 10 & 5 \\ 5 & 0 & 10 \\ 5 & 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

An industrial production problem.

Goal: To find values of $x_i, i = 1,2,3$ that satisfy the model equations.

Model:

$$\begin{array}{rrcr} 15x_1 & +20x_2 & +10x_3 & = 80 \\ 5x_1 & +10x_2 & +5x_3 & = 30 \\ 10x_1 & +10x_2 & +5x_3 & = 40 \\ 5x_1 & +0x_2 & +10x_3 & = 45 \\ 5x_1 & +0x_2 & +5x_3 & = 30 \end{array}$$

- Translate this problem into the language of matrices and vectors.

Let: $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 80 \\ 30 \\ 40 \\ 45 \\ 30 \end{bmatrix}$

$$A = \begin{bmatrix} 15 & 20 & 10 \\ 5 & 10 & 5 \\ 10 & 10 & 5 \\ 5 & 0 & 10 \\ 5 & 0 & 5 \end{bmatrix}$$

Model: $A\vec{x} = \vec{b}$

Goal: Solve for the $\vec{x}$ that satisfies the model equation.

A stochastic weather prediction problem.

- If it is sunny one day, there is a 50% chance it will be sunny the next day, a 40% chance that it will be cloudy the next day, and a 10% chance that it will rain the next day.
- If it is cloudy one day, there is a 50% chance it will remain cloudy the next day, and a 25% chance each of raining or being sunny the next day.
- If it is raining one day, there is a 40% chance it will still be raining the next day, a 40% chance it will be cloudy, and a 20% chance it will be sunny.
- If it is sunny on Wednesday, what is the probability that it will be raining on Sunday?
- What is the probability it will be sunny on a randomly selected day of the year?

- Let $p_{X,i}$ be the probability of being in state “X” on day “i”, where $X=S$ for sunny, $X=C$ for cloudy and $X=R$ for raining.

$$\begin{aligned}p_{S,i+1} &= 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i} \\p_{C,i+1} &= 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i} \\p_{R,i+1} &= 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i}\end{aligned}$$

- **Problem 1:** We want to find $p_{R,4}$, given that $p_{S,0} = 1$ (and $p_{C,0} = p_{R,0} = 0$).
- **Problem 2:** Solve the system to have the same probabilities on days i and $i+1$; this is the “equilibrium distribution”.
- **Translate this problem into the language of matrices and vectors.**

A stochastic weather prediction problem.

Variables are the same in every equation → Vector.

Coefficients vary in both row and column → Matrix.

$$\begin{bmatrix} 0.50p_{S,i} & +0.25p_{C,i} & +0.20p_{R,i} \\ 0.40p_{S,i} & +0.50p_{C,i} & +0.40p_{R,i} \\ 0.10p_{S,i} & +0.25p_{C,i} & +0.40p_{R,i} \end{bmatrix} = \begin{bmatrix} 0.50 & 0.25 & 0.20 \\ 0.40 & 0.50 & 0.40 \\ 0.10 & 0.25 & 0.40 \end{bmatrix} \begin{bmatrix} p_{S,i} \\ p_{C,i} \\ p_{R,i} \end{bmatrix}$$

A stochastic weather prediction problem.

- **Goal 1:** We want to find $p_{R,4}$, given that $p_{S,0} = 1$ (and $p_{C,0} = p_{R,0} = 0$).
- **Goal 2:** Solve the system to have the same probabilities on days i and $i+1$.

Model:

$$\begin{aligned} p_{S,i+1} &= 0.50p_{S,i} + 0.25p_{C,i} + 0.20p_{R,i} \\ p_{C,i+1} &= 0.40p_{S,i} + 0.50p_{C,i} + 0.40p_{R,i} \\ p_{R,i+1} &= 0.10p_{S,i} + 0.25p_{C,i} + 0.40p_{R,i} \end{aligned}$$

- **Translate this problem into the language of matrices and vectors.**

Let: $\vec{p}_i = \begin{bmatrix} p_{S,i} \\ p_{C,i} \\ p_{R,i} \end{bmatrix}, \vec{p}_{i+1} = \begin{bmatrix} p_{S,i+1} \\ p_{C,i+1} \\ p_{R,i+1} \end{bmatrix}$

$$A = \begin{bmatrix} 0.50 & 0.25 & 0.20 \\ 0.40 & 0.50 & 0.40 \\ 0.10 & 0.25 & 0.40 \end{bmatrix}$$

Model: $\vec{p}_{i+1} = A\vec{p}_i$

Goal 1: Find $\vec{p}_4$ (element 3) given $\vec{p}_0$.

Goal 2: Find $\vec{p}^*$ such that $\vec{p}^* = A\vec{p}^*$.

An economic supply and demand problem.

- To produce and distribute \$1M worth of energy requires \$0.1M worth of metals, \$0.3 M worth of transportation, \$0.1M worth of construction materials, and \$0.3M worth of input energy.
 - To produce and distribute \$1M worth of metals requires \$0.3M of energy, \$0.2M of transportation, \$0.2M of construction materials, and \$0.1M of metals.
 - To provide \$1M worth of transportation requires \$0.2M of energy, \$0.5M of metals and \$0.2M of construction materials.
 - To produce and distribute \$1M of construction materials requires \$0.2M of energy, \$0.1M of construction materials, \$0.4M of transportation and \$0.1M of metals.
 - If consumer demand requires \$80M of energy, \$50M of construction and \$100M of transportation, how much must each industry produce?
 - Let x_i be the total supply (in millions of dollars produced) by economic sector i .
 - Let d_i be the total demand (in millions of dollars needed) for the output of economic sector i .
 - Let the sectors be defined as: energy $i=1$; metals $i=2$; transportation $i=3$; and construction materials $i=4$.
- $$\begin{aligned}d_1 &= 0.3x_1 + 0.3x_2 + 0.2x_3 + 0.2x_4 + 80 \\d_2 &= 0.1x_1 + 0.1x_2 + 0.5x_3 + 0.1x_4 + 0 \\d_3 &= 0.3x_1 + 0.2x_2 + 0.0x_3 + 0.4x_4 + 100 \\d_4 &= 0.1x_1 + 0.2x_2 + 0.2x_3 + 0.1x_4 + 50\end{aligned}$$
- We would like to find a consistent solution, where the total supply is equal to the total demand for each sector of the economy.
- $$d_i = x_i, \forall i$$
- Translate this problem into the language of matrices and vectors.

An economic supply and demand problem.

First try:

Variables are the same in every equation → Vector.

Coefficients vary in both row and column → Matrix.

But what do we do about the last column?

$$\begin{bmatrix} 0.3x_1 & +0.3x_2 & +0.2x_3 & +0.2x_4 & +80 \\ 0.1x_1 & +0.1x_2 & +0.5x_3 & +0.1x_4 & +0 \\ 0.3x_1 & +0.2x_2 & +0.0x_3 & +0.4x_4 & +100 \\ 0.1x_1 & +0.2x_2 & +0.2x_3 & +0.1x_4 & +50 \end{bmatrix} = \begin{bmatrix} 0.3 & 0.3 & 0.2 & 0.2 & 80 \\ 0.1 & 0.1 & 0.5 & 0.1 & 0 \\ 0.3 & 0.2 & 0.0 & 0.4 & 100 \\ 0.1 & 0.2 & 0.2 & 0.1 & 50 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ 1 \end{bmatrix}$$

- This is formally correct but looks a bit odd; the vector isn't just variables!

An economic supply and demand problem.

Second try:

Split the system into a piece with variables and a piece without.

$$\begin{bmatrix} 0.3x_1 & +0.3x_2 & +0.2x_3 & +0.2x_4 & +80 \\ 0.1x_1 & +0.1x_2 & +0.5x_3 & +0.1x_4 & +0 \\ 0.3x_1 & +0.2x_2 & +0.0x_3 & +0.4x_4 & +100 \\ 0.1x_1 & +0.2x_2 & +0.2x_3 & +0.1x_4 & +50 \end{bmatrix} = \begin{bmatrix} 0.3x_1 & +0.3x_2 & +0.2x_3 & +0.2x_4 \\ 0.1x_1 & +0.1x_2 & +0.5x_3 & +0.1x_4 \\ 0.3x_1 & +0.2x_2 & +0.0x_3 & +0.4x_4 \\ 0.1x_1 & +0.2x_2 & +0.2x_3 & +0.1x_4 \end{bmatrix} + \begin{bmatrix} 80 \\ 0 \\ 100 \\ 50 \end{bmatrix}$$

- **Note, all of these objects are vectors. The elements of the vectors are scalar sums.**

An economic supply and demand problem.

Second try:

Then work with just the first component.

$$\begin{bmatrix} 0.3x_1 & +0.3x_2 & +0.2x_3 & +0.2x_4 \\ 0.1x_1 & +0.1x_2 & +0.5x_3 & +0.1x_4 \\ 0.3x_1 & +0.2x_2 & +0.0x_3 & +0.4x_4 \\ 0.1x_1 & +0.2x_2 & +0.2x_3 & +0.1x_4 \end{bmatrix} = \begin{bmatrix} 0.3 & 0.3 & 0.2 & 0.2 \\ 0.1 & 0.1 & 0.5 & 0.1 \\ 0.3 & 0.2 & 0.0 & 0.4 \\ 0.1 & 0.2 & 0.2 & 0.1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

- This now looks like our standard form.

An economic supply and demand problem.

Goal: to solve for $d_i = x_i$, for all $i = 1, 2, 3, 4$.

Model:

$$\begin{array}{rclclcl} d_1 = & 0.3x_1 & +0.3x_2 & +0.2x_3 & +0.2x_4 & +80 \\ d_2 = & 0.1x_1 & +0.1x_2 & +0.5x_3 & +0.1x_4 & +0 \\ d_3 = & 0.3x_1 & +0.2x_2 & +0.0x_3 & +0.4x_4 & +100 \\ d_4 = & 0.1x_1 & +0.2x_2 & +0.2x_3 & +0.1x_4 & +50 \end{array}$$

- Translate this problem into the language of matrices and vectors.

Let: $\vec{d} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{b} = \begin{bmatrix} 80 \\ 0 \\ 100 \\ 50 \end{bmatrix}$

$$A = \begin{bmatrix} 0.3 & 0.3 & 0.2 & 0.2 \\ 0.1 & 0.1 & 0.5 & 0.1 \\ 0.3 & 0.2 & 0.0 & 0.4 \\ 0.1 & 0.2 & 0.2 & 0.1 \end{bmatrix}$$

Model: $\vec{d} = A\vec{x} + \vec{b}$

Goal: Solve for $\vec{d} = \vec{x}$.

A constrained optimization problem.

- A farmer must decide how to divide 200 acres of arable land between two major crops (wheat and corn).
- Corn is more costly to produce, requiring 4 worker-days of labor and \$200 of capital for each acre planted;
- Wheat requires only a single worker-day of labor and \$100 of capital per acre;
- However, corn produces a profit of \$600/acre, while wheat produces a profit of only \$400/acre.
- If the farmer has \$22,000 of available capital and 320 worker-days of labor, what is the most profitable planting strategy?

Let x be the number of acres of corn, y be the number of acres of wheat, and P be the total profit.

Maximize:

$$P = 600x + 400y$$

Subject to the constraints:

$$x + y \leq 200$$

$$4x + y \leq 320$$

$$200x + 100y \leq 22,000$$

Note that the last constraint can be simplified to:

$$2x + 1y \leq 220$$

- We wish to find the values of x and y that will give us the largest possible profit, P , while satisfying all the constraints.
- Translate this problem into the language of matrices and vectors.

A constrained optimization problem.

Two parts: Target function and constraints.

Target function: $[600x + 400y] = [600 \quad 400] \begin{bmatrix} x \\ y \end{bmatrix}$

Constraints: $\begin{bmatrix} x & +y \\ 4x & +y \\ 2x & +y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

A constrained optimization problem.

Goal:

Find the values of x and y that maximize:

$$P = 600x + 400y.$$

subject to the constraints:

$$x + y \leq 200$$

$$4x + y \leq 320$$

$$2x + y \leq 220$$

- Translate this problem into the language of matrices and vectors.

Let: $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \vec{b} = \begin{bmatrix} 200 \\ 320 \\ 220 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \\ 2 & 1 \end{bmatrix}, M = [600 \quad 400]$$

Goal:

Find the $\vec{x}$ that maximizes:

$$P = M\vec{x}$$

Subject to the constraints:

$$A\vec{x} \leq \vec{b}$$

Unit 2.4

Abstract (symbolic) matrix algebra.

Symbolic matrix algebra.

- Just as we can manipulate algebraic expressions of scalar variables and constants, we can do the same with expressions of matrices and vectors:

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

$$x^2 + 6x + 8 = (x^2 + 6x + 9) - 1 = (x + 3)^2 - 1$$

- **We need to understand the rules!**

Rules of matrix algebra.

- Matrix addition is associative and commutative:

$$A + (B + C) = (A + B) + C$$

$$A + B = B + A$$

- **Matrix multiplication is associative, but usually NOT commutative:**

$$A(BC) = (AB)C$$

$$AB \neq BA$$

- Matrix multiplication and addition are distributive:

$$A(B + C) = AB + AC$$

$$(A + B)C = AC + BC$$

- Treat vectors as (1 x n) or (n x 1) matrices and obey all the same rules!

Rules of matrix algebra.

- **Scalar** multiplication is associative and commutative:

$$k(A + B) = kA + kB$$

$$(k + l)A = kA + lA$$

$$kA = Ak$$

$$k(AB) = (kA)B = (Ak)B = A(kB) = A(Bk) = (AB)k$$

- **All these rules assume that the dimensions match in an appropriate way; the rules only hold when the expression makes sense.**
- Treat vectors as (1 x n) or (n x 1) matrices and obey all the same rules!

Rules of matrix algebra.

- The transpose of a matrix can be distributed into matrix addition and matrix multiplication.:

$$\begin{aligned}(A + B)^T &= A^T + B^T \\ (AB)^T &= B^T A^T\end{aligned}$$

- Note the change of order when the transpose is distributed into a product; for non-square matrices, the sizes wouldn't match if we did not do this.

Rules of matrix algebra.

- The inverse of a matrix can be distributed into matrix multiplication but not matrix addition and matrix multiplication:

$$(A + B)^{-1} \neq A^{-1} + B^{-1}$$
$$(AB)^{-1} = B^{-1}A^{-1}$$

- **NOTE:** The matrix inverse does not distribute into matrix addition, the inverse may not even exist for all the matrices.
- On the other hand, if AB has an inverse, then so do A and B , and vice versa.

Rules of matrix algebra.

- Remember the additive and multiplicative identities:

$$\mathbf{0} + A = A + \mathbf{0} = A$$

$$IA = AI = A$$

$$\mathbf{0}A = \mathbf{0}, \quad A\mathbf{0} = \mathbf{0}$$

- NOTE:** The product of a zero with any matrix is also a zero matrix, but the size of the result depends on the dimensions of the zero matrix. The product of a matrix with the scalar 0 is a zero matrix of the same size.
- You may sometimes encounter the **ones matrix** (a matrix with every element equal to 1). The ones matrix doesn't really have any special algebraic rules; importantly, it is NOT the identity:

$$\mathbf{1}A \neq A, \quad A\mathbf{1} \neq A, \quad \mathbf{1}A \neq A\mathbf{1}$$

Unit 2.4.1

**Worked problems in symbolic
matrix algebra.**

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- **What properties must each statement have to even make sense.**

$$A + B + (C + B) + C = A + 2B + 2C$$

$$2(A + B) - (C + 2B) = 2(A - C)$$

$$(X + Y)^2 = X^2 + 2XY + Y^2$$

$$\left((X + Y)(X - Y)\right)^T + (XY)^T = (X^T)^2 + X^T Y^T - (Y^T)^2$$

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- What properties must each statement have to even make sense.

$$A + B + (C + B) + C = A + 2B + 2C$$

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- What properties must each statement have to even make sense.

$$A + B + (C + B) + C = A + 2B + 2C$$

- **Matrices A, B and C must all be of the same size.**
- **The statement is TRUE:**

$$\begin{aligned} A + B + (C + B) + C &= A + B + (C + B) + C \\ &= A + (B + B) + (C + C) \\ &= A + 2B + 2C \end{aligned}$$

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- What properties must each statement have to even make sense.

$$2(A + B) - (C + 2B) = 2(A - C)$$

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- What properties must each statement have to even make sense.

$$2(A + B) - (C + 2B) = 2(A - C)$$

- **Matrices A, B and C must all be of the same size.**
- **The statement is FALSE (unless C=0):**

$$\begin{aligned} 2(A + B) - (C + 2B) &= 2A + 2B - C - 2B \\ &= 2A - C \\ &\neq 2(A - C) \end{aligned}$$

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- What properties must each statement have to even make sense.

$$(X + Y)^2 = X^2 + 2XY + Y^2$$

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- What properties must each statement have to even make sense.

$$(X + Y)^2 = X^2 + 2XY + Y^2$$

- **Matrices X and Y must be square matrices of the same size.**
- **The statement is FALSE (unless $XY=YX$, commuting matrices):**

$$\begin{aligned}(X + Y)^2 &= (X + Y)(X + Y) \\ &= XX + XY + YX + YY \\ &= X^2 + XY + YX + Y^2 \\ &\neq X^2 + 2XY + Y^2\end{aligned}$$

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- What properties must each statement have to even make sense.

$$\left((X + Y)(X - Y)\right)^T + (XY)^T = (X^T)^2 + X^T Y^T - (Y^T)^2$$

Symbolic matrix algebra practice.

- Determine whether each of the following equalities is true or false.
- What properties must each statement have to even make sense.

$$\left((X + Y)(X - Y)\right)^T + (XY)^T = (X^T)^2 + X^T Y^T - (Y^T)^2$$

- **Matrices X and Y must be square matrices of the same size.**
- **The statement is TRUE:**

$$\begin{aligned}\left((X + Y)(X - Y)\right)^T + (XY)^T &= (X - Y)^T (X + Y)^T + Y^T X^T \\ &= (X^T - Y^T)(X^T + Y^T) + Y^T X^T \\ &= X^T X^T + X^T Y^T - Y^T X^T - Y^T Y^T + Y^T X^T \\ &= (X^T)^2 + X^T Y^T - (Y^T)^2\end{aligned}$$